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COMP-2024 / COMP-2039

Artificial Intelligence Methods

Lecture 5

Evolutionary Algorithms I

Learning Outcomes

IDENTIFY

1. Evolutionary Algorithms
2. Genetic Algorithms (GAs)
3. Memetic Algorithms
4. **Benchmark Functions**

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Benchmark Functions

Benchmark functions

- **Benchmark (test) functions** for optimization are standardized mathematical problems used to evaluate the performance of optimization algorithms.
- **Benchmark functions** serves as a testbed for performance comparison of (meta/hyper)heuristic optimisation algorithms.
- These functions are used to test algorithms for various features, including convergence rate, handling of local optima, scalability, and robustness.
- They play a crucial role in the evaluation and comparison of optimization algorithms, especially in the context of metaheuristic and hyper-heuristic approaches.

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Why to use benchmark (test) functions for optimisation

- **Performance Comparison**
  - **Purpose**: Benchmark functions provide a standardized set of problems that can be used to assess the performance of different optimization algorithms.
  - **Objective**: By using the same set of functions, researchers and practitioners can effectively compare the efficiency, convergence speed, and robustness of various algorithms.

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Why to use benchmark (test) functions for optimisation

- **Known Global Minimum**
  - **Assurance**: Each benchmark function has a well-defined global minimum that is known in advance.
  - **Importance**: This allows for a clear assessment of how close the optimization algorithm can get to the optimal solution, enabling a quantifiable measure of success.

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Why to use benchmark (test) functions for optimisation

- **Ease of Computation**
  - **Simplicity**: Benchmark functions are designed to be easily computable, allowing for quick evaluations.
  - **Efficiency**: This ease of computation is crucial for testing algorithms across numerous iterations or across different parameter settings without excessive computational overhead.

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Why to use benchmark (test) functions for optimisation

- **Variety of Problems**: Each benchmark function exhibits unique characteristics, such as:
  - **Number of local minima**: Some functions have many local minima, making them complex and challenging for optimization algorithms.
  - **Smoothness**: Some are smooth and continuous, while others may have discontinuities or steep gradients.
  - **Dimensionality**: Functions can vary in the number of dimensions, testing the algorithm's scalability.
- **Real-World Representation**: These characteristics can be reflective of different real-world optimization problems, making benchmarks relevant for practical applications.

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Example of Benchmarking Functions for Continuous Optimization Problems

No.	Name	*D*	*C*	*S*	Function Definition	f_{min}
f_1	Sphere	10,30,50	U	$[-5.12, 5.12]^D$	$f_1(x) = \sum_{i=1}^{d} x_i^2$	0.0
f_2	Step	10,30,50	U	$[-100, 100]^D$	$f_2(x) = \sum_{i=1}^{d}(\lfloor x_i + 0.5 \rfloor^2)$	0.0
f_3	Zakharov	10,30,50	U	$[-5, 10]^D$	$f_3(x) = \sum_{i=1}^{d} x_i^2 + \left(\sum_{i=1}^{d} 0.5ix_i\right)^2 + \left(\sum_{i=1}^{d} 0.5ix_i\right)^4$	0.0
f_4	Griewangk	10,30,50	M	$[-600, 600]^D$	$f_4(x) = \frac{1}{4000}\sum_{i=1}^{d} x_i^2 - \prod_{i=1}^{d} \cos\frac{x_i}{\sqrt{i}} + 1$	0.0
f_5	Rastrigin	10,30,50	M	$[-15, 15]^D$	$f_5(x) = \sum_{i=1}^{d}\left[x_i^2 - 10\cos(2\pi x_i) + 10\right]$	0.0
f_6	Rosenbrock	10,30,50	U	$[-15, 15]^D$	$f_6(x) = \sum_{i=1}^{d-1}\left[100\left(x_{i+1} - x_i^2\right)^2 + (x_i - 1)^2\right]$	0.0
f_7	Ackley	10,30,50	M	$[-32, 32]^D$	$f_7(x) = -20exp\left(-0.2\sqrt{\frac{1}{d}\sum_{i=1}^{d} x_i^2}\right) - exp\left(\frac{1}{n}\sum_{i=1}^{n}\cos 2\pi x_i\right) + 20 + e$	0.0
f_8	Schwefel	10,30,50	M	$[-500, 500]^D$	$f_8(x) = 418.9829 * d - \sum_{i=1}^{n} -x_i \sin\left(\sqrt{\lvert x_i \rvert}\right)$	0.0
f_9	Easom	10,30,50	M	$[-2\pi, 2\pi]^D$	$f_9(x) = -(-1)^d * \left(\prod_1^d cos^2(x_i)\right) * exp\left[\sum_1^d (x_i - \pi)^2\right]$	-1.000
f_{10}	Michalewicz	10,30,50	M	$[0, \pi]^D$	$f_{10}(x) = -\sum_{i=1}^{d} \sin(x_i) sin^{2m}\left(\frac{ix_i^2}{\pi}\right)$	NA

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Classification of benchmark (test) functions

A common classification is:

- Continuity (Differentiability)
  - Discontinuous vs continuous
- Dimensionality
  - Scalability
- Modality
  - Unimodal
  - Multimodal with few local minima
  - Multimodal with exponential number of local minima
- Separability $\frac{\partial f(\bar{x})}{\partial x_i} = g(x_i)h(\bar{x})$

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A collection of **unconstrained** benchmark functions: https://arxiv.org/pdf/1308.4008.pdf

Continuity (Differentiability)

- **Discontinuous Function**
- **Example**: Step Function
  - **Characteristics**: Abrupt changes at integer values; non-differentiable.

- **Continuous Function**
- **Example**: Rosenbrock Function
  - **Characteristics**: Smooth and continuous; differentiable everywhere.

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Dimensionality

- **Low-Dimensional Function**
- **Example**: Sphere Function (2D)
  - **Characteristics**: Simple, easy to optimize.

- **High-Dimensional Function**
- **Example**: Rastrigin Function (n-D)
  - **Characteristics**: Highly multimodal; complexity increases with dimensions.

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Separability

- **Separable Function**
- **Example**: Separable Quadratic Function
  - **Characteristics**: Variables can be optimized independently

- **Non-Separable Function**
- **Example**: Rosenbrock Function
  - **Characteristics**: Interdependent variables; optimization is more complex due to the interaction between dimensions.

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Modality

- **Unimodal Function**
  - **Example**: Quadratic Function
  - **Characteristics**: Only one global minimum

- **Multimodal Functions**
  - **With Few Local Minima**: Schwefel Function
  - **Characteristics**: Multiple local minima, but manageable.
  - **With Exponential Number of Local Minima:** Rastrigin Function
  - **Characteristics**: Exhibits a high number of local minima, making it very challenging.

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Unimodal Functions

- Used to test the ability of an algorithm to converge to a single global minimum.

- **Sphere Function**:

$$f(\mathbf{x}) = \sum_{i=1}^{n} x_i^2$$

  - Domain: $x_i \in [-5.12, 5.12]$
  - Global Minimum: $f(\mathbf{x}) = 0$ at $\mathbf{x} = \mathbf{0}$

- **Rosenbrock Function (Banana Function)**:

$$f(\mathbf{x}) = \sum_{i=1}^{n-1} \left[100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2\right]$$

  - Domain: $x_i \in [-2.048, 2.048]$
  - Global Minimum: $f(\mathbf{x}) = 0$ at $\mathbf{x} = [1, 1, \ldots, 1]$

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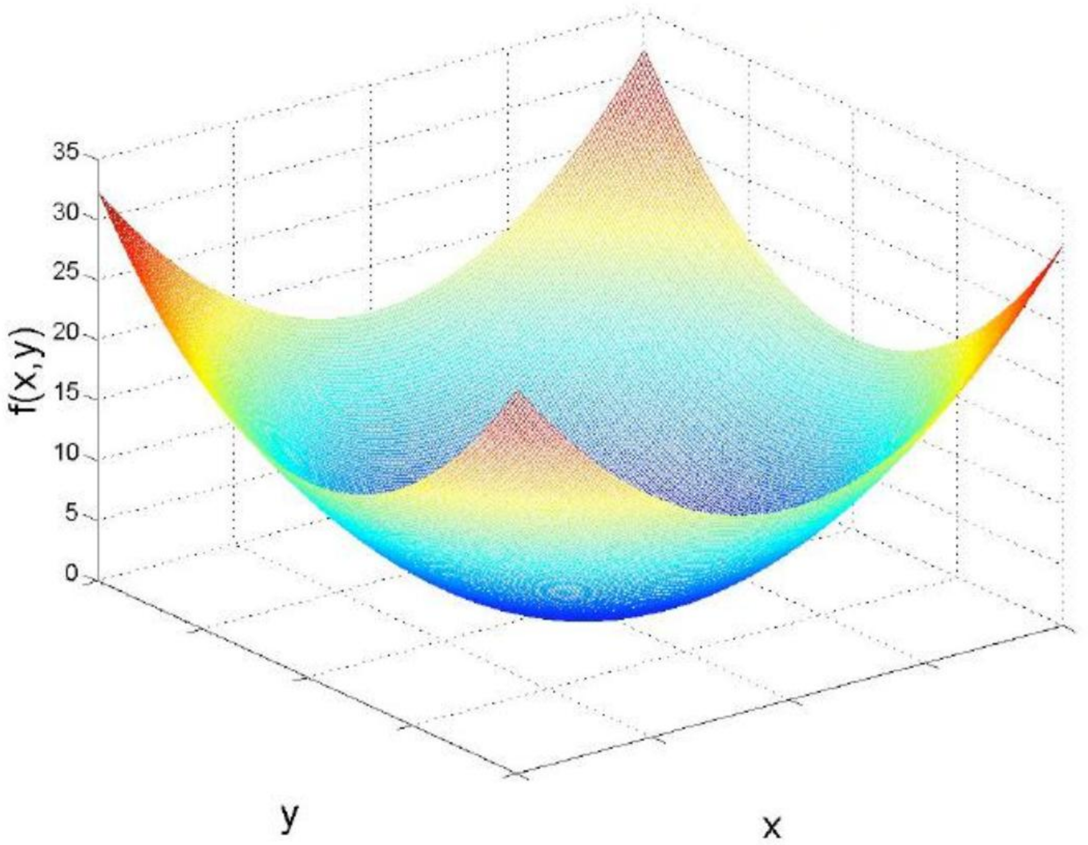
Example – unimodal function Sphere Function

$$f(x) = \sum_{i=1}^{n} x_i^2$$

$$f(x_1, \dots, x_n) = f(0, \dots, 0) = 0$$



- Continuous
- Differentiable
- Separable
- Scalable

$n = 2$

$$-5.12 \le x, y \le 5.12$$

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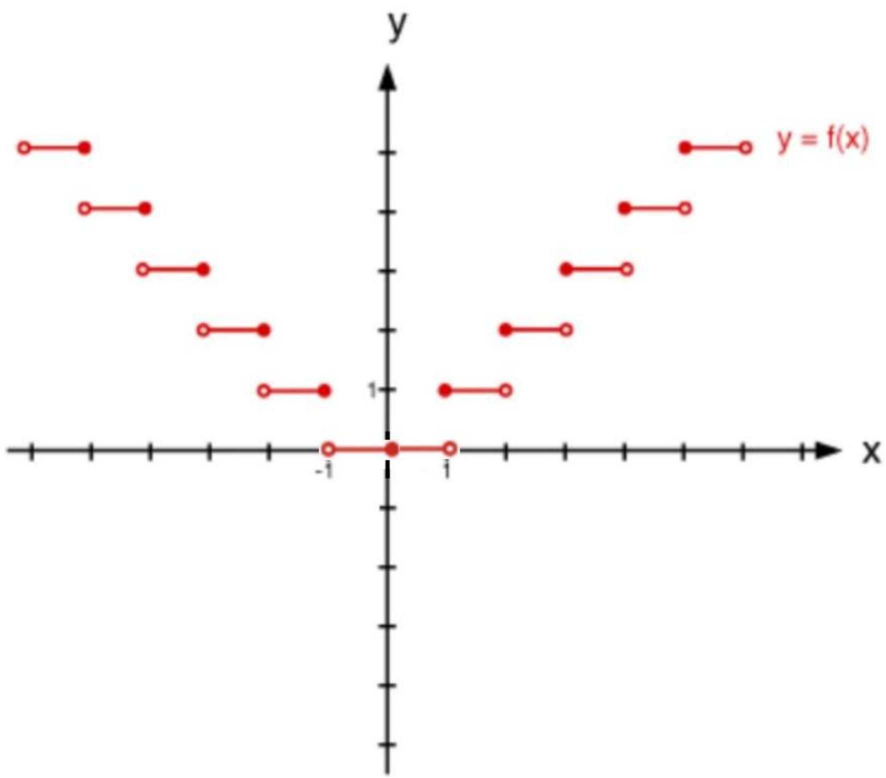
Example – unimodal function Step Function

$$f(x) = \sum_{i=1}^{n} \|x_i\|$$

$$f(x_1, \dots, x_n) = f(0, \dots, 0) = 0$$



- Discontinuous
- Non-differentiable
- Separable
- Scalable

$n = 2$

$$-100 \leq x_i \leq 100$$

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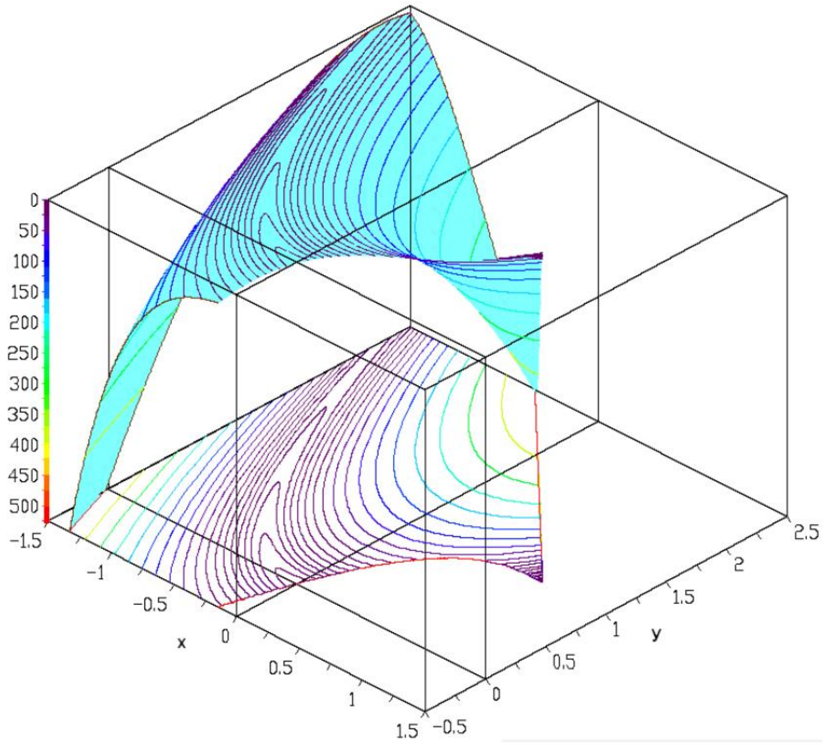
Rosenbrock function constrained with a cubic and a line

$$f(x,y) = (1-x)^2 + 100(y-x^2)^2$$

$$Subjected\ to:\ (x-1)^3 - y + 1 < 0\ and\ x + 2\ - 2 < 0$$

$$f(1.0, 1.0) = 1.0$$

$$-1.5 \le x \le 1.5$$
$$-0.5 \le y \le 0.5$$



- Continuous
- Differentiable
- Non- separable
- Non-scalable

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Separable Functions

- The variables can be optimized independently.

1. Sphere Function

$$f(\mathbf{x}) = \sum_{i=1}^{n} x_i^2$$

- **Domain**: $x_i \in [-5.12, 5.12]$
- **Global Minimum**: $f(\mathbf{x}) = 0$ at $\mathbf{x} = \mathbf{0}$
- **Properties**:
  - Unimodal and convex.
  - Each variable contributes independently to the objective value.

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What Does "Domain" Mean in a Benchmarking Function?

In the context of optimization benchmarking functions, the **domain** refers to the range of permissible values for the decision variables $\mathbf{x} = [x_1, x_2, \ldots, x_n]$. It defines the **search space** where the optimization algorithm is allowed to operate. For each variable x_i, the domain is typically specified as an interval $[a_i, b_i]$, where:

- a_i: The lower bound for x_i.
- b_i: The upper bound for x_i.

For example, in the **Sphere Function**, the domain might be $x_i \in [-5.12, 5.12]$, meaning each variable x_i can take values between -5.12 and 5.12.

- The domains in benchmarking functions are often chosen based on practical considerations, theoretical properties, or historical precedent

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Examples of Domain Selection

Function	Domain (x_i)	Reason
Sphere Function	$[-5.12, 5.12]$	Historical precedent; covers key features.
Rastrigin Function	$[-5.12, 5.12]$	Challenging due to multimodality within range.
Ackley Function	$[-32.768, 32.768]$	Captures complex landscapes with steep regions.
Rosenbrock Function	$[-2.048, 2.048]$	Simpler optimization problem within bounds.
Schwefel Function	$[-500, 500]$	Reflects large-scale landscapes.

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Delta Evaluation

- **Delta evaluation** is a technique used in optimization, particularly when dealing with separable functions, to efficiently assess changes in the function's value based on small adjustments to its input variables.
- This method focuses on how **small changes (deltas)** in the input variables affect the overall function value. It simplifies the optimization process by allowing changes to be evaluated independently.

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Delta Evaluation in Function Optimisation

- Separable functions allows delta evaluation
- Example, sphere function $f(x) = \sum_{i=1}^{n} x_i^2$

$\underline{S}$		X_{43}				$\underline{S'}$		X_{43}		
x_1	…	6	…	x_n	$\Rightarrow$	x_1	…	5	…	x_n

$$f(s) = \sum \qquad \Rightarrow \qquad f(s') = \sum + \boldsymbol{\Delta}$$

$$(5^2 - 6^2)$$

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Delta Evaluation in Function Optimisation Example: Sphere Function

The **Sphere Function** is a commonly used benchmark in optimization. Its formula is:

$$f(x) = \sum_{i=1}^{n} x_i^2$$

Where n is the number of dimensions (variables).

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Delta Evaluation in Function Optimisation Example: Sphere Function

1. **Initial Values**: Suppose we have initial values for n dimensions:
   - $x_1 = 6$
   - $x_2 = x_3 = \ldots = x_n$ (for simplicity, assume all other dimensions are the same).
2. **Adjusting One Variable**: Let's say we adjust x_1 to a new value $x_1' = 5$ while keeping the others unchanged.
3. **Function Evaluation**:
   - **Original Function Value**:

$$f(x) = x_1^2 + x_2^2 + \ldots + x_n^2$$

   - **New Function Value** after the change:

$$f(s') = (x_1')^2 + x_2^2 + \ldots + x_n^2$$

4. **Delta Calculation**:
   - The change in the function value (Δf) can be calculated as:

$$\Delta f = f(s') - f(s)$$

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Delta Evaluation in Function Optimisation Example: Sphere Function

5. **Example Calculation**:

- If $n = 3$ for simplicity:
  - Original values: $x_1 = 6$, $x_2 = 6$, $x_3 = 6$
  - Original function value:

$$f(s) = 6^2 + 6^2 + 6^2 = 36 + 36 + 36 = 108$$

  - New value for x_1: $x_1' = 5$
  - New function value:

$$f(s') = 5^2 + 6^2 + 6^2 = 25 + 36 + 36 = 97$$

  - Change in function value:

$$\Delta f = 97 - 108 = -11$$

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Non-Separable Functions

- The variables interact, making them harder to optimize.

  - **Griewank Function:**

$$f(\mathbf{x}) = 1 + \frac{1}{4000}\sum_{i=1}^{n} x_i^2 - \prod_{i=1}^{n} \cos\left(\frac{x_i}{\sqrt{i}}\right)$$

  - Domain: $x_i \in [-600, 600]$
  - Global Minimum: $f(\mathbf{x}) = 0$ at $\mathbf{x} = \mathbf{0}$

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Multimodal Functions

- Used to test an algorithm's ability to escape local optima and find the global optimum.
  - **Rastrigin Function**:

$$f(\mathbf{x}) = 10n + \sum_{i=1}^{n} \left[x_i^2 - 10\cos(2\pi x_i)\right]$$

    - Domain: $x_i \in [-5.12, 5.12]$
    - Global Minimum: $f(\mathbf{x}) = 0$ at $\mathbf{x} = \mathbf{0}$
  - **Ackley Function**:

$$f(\mathbf{x}) = -20\exp\left(-0.2\sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2}\right) - \exp\left(\frac{1}{n}\sum_{i=1}^{n}\cos(2\pi x_i)\right) + 20 + e$$

    - Domain: $x_i \in [-32.768, 32.768]$
    - Global Minimum: $f(\mathbf{x}) = 0$ at $\mathbf{x} = \mathbf{0}$

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Example – multimodal function Rastrigin's Function

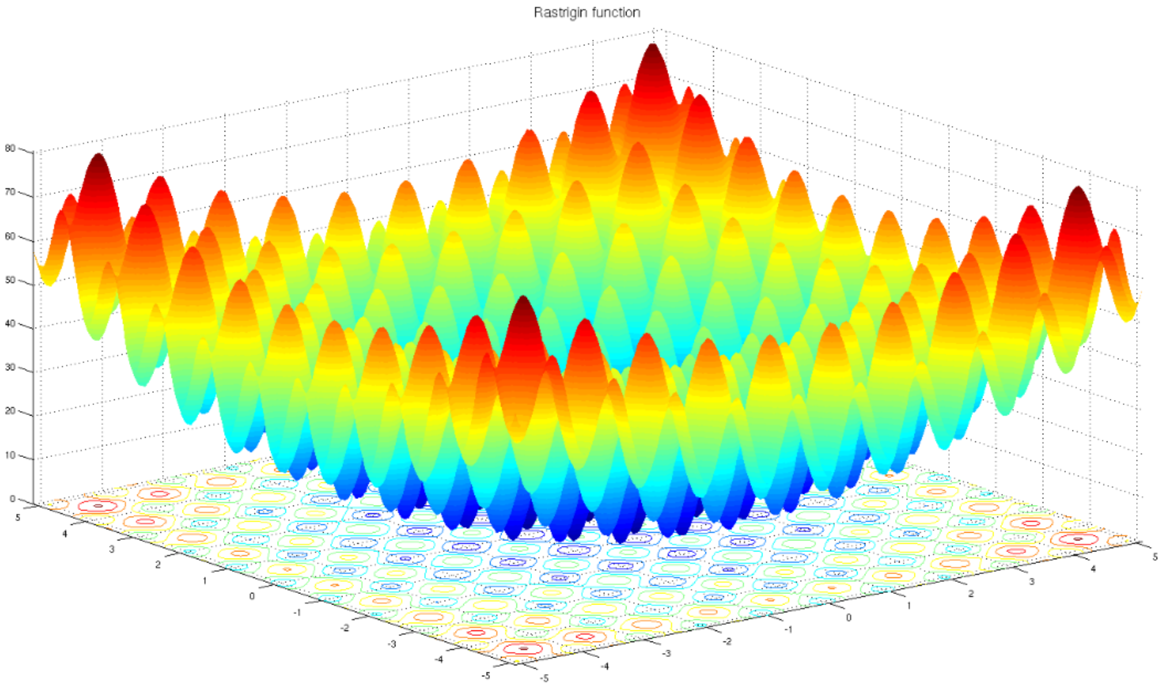
$$f(x) = An + \sum_{i=1}^{n}[x_i^2 - A\cos(2\pi x_i)]$$

where: A = 10

$n = 2$

$$f(0,0) = 0$$

$$-5.12 \leq x, y \leq 5.12$$



- Continuous
- Differentiable
- Separable
- Scalable

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Example – multimodal function Ackley's Function

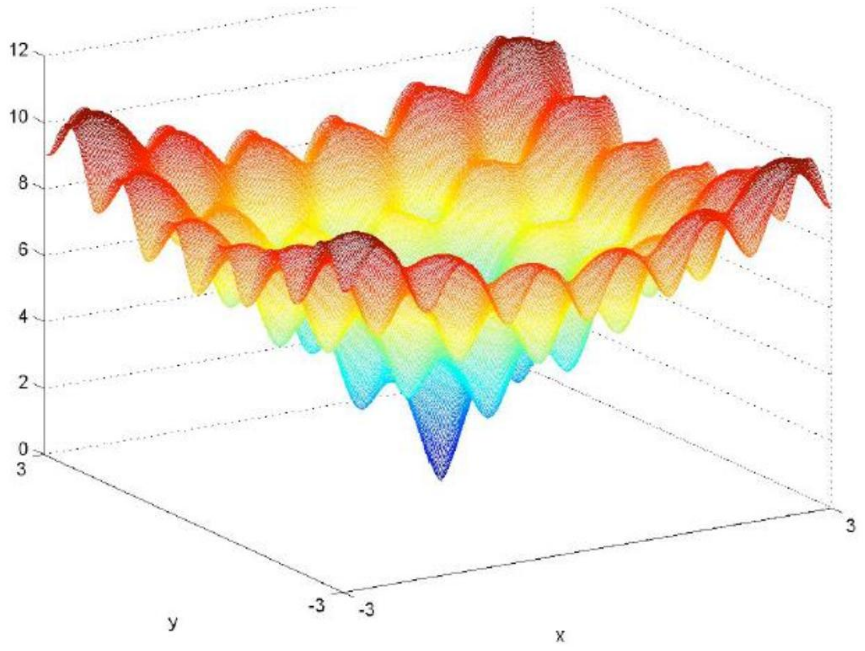
$$f(x,y) = -20\exp\left(-0.2\sqrt{0.5(x^2+y^2)}\right)$$

$$-\exp\left(0.5(\cos(2\pi x)+\cos(2\pi y))\right) + e + 20$$

$n = 2$

$$f(0,0) = 0$$

$$-5 \le x, y \le 5$$



- Continuous
- Differentiable
- Non- separable
- Non-scalable

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Criteria for Choosing Benchmark Functions

- **Scalability**: Functions should allow testing for different dimensions n.
- **Diversity**: Include unimodal, multimodal, separable, non-separable, and noisy functions.
- **Realism**: Functions like composite benchmarks (e.g., hybrid or rotated functions) mimic real-world complexities.
- **Known Global Optimum**: Helps evaluate the accuracy of the algorithm.

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Benchmark Suites

- COCO (Comparing Continuous Optimizers)
- BBOB (Black-Box Optimization Benchmarking)
- CEC (Congress on Evolutionary Computation)
- Test Problems for Constrained Optimization
- Hyperparameter Optimization Benchmarks
- etc.

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COCO: COmparing Continuous Optimizers

- **COCO (Comparing Continuous Optimizers)** is one of the popular benchmark suites used in the optimization community. It is specifically designed for benchmarking optimization algorithms on **continuous black-box optimization problems**.
- COCO provides a framework for testing, analyzing, and comparing optimization algorithms on standardized problem sets.

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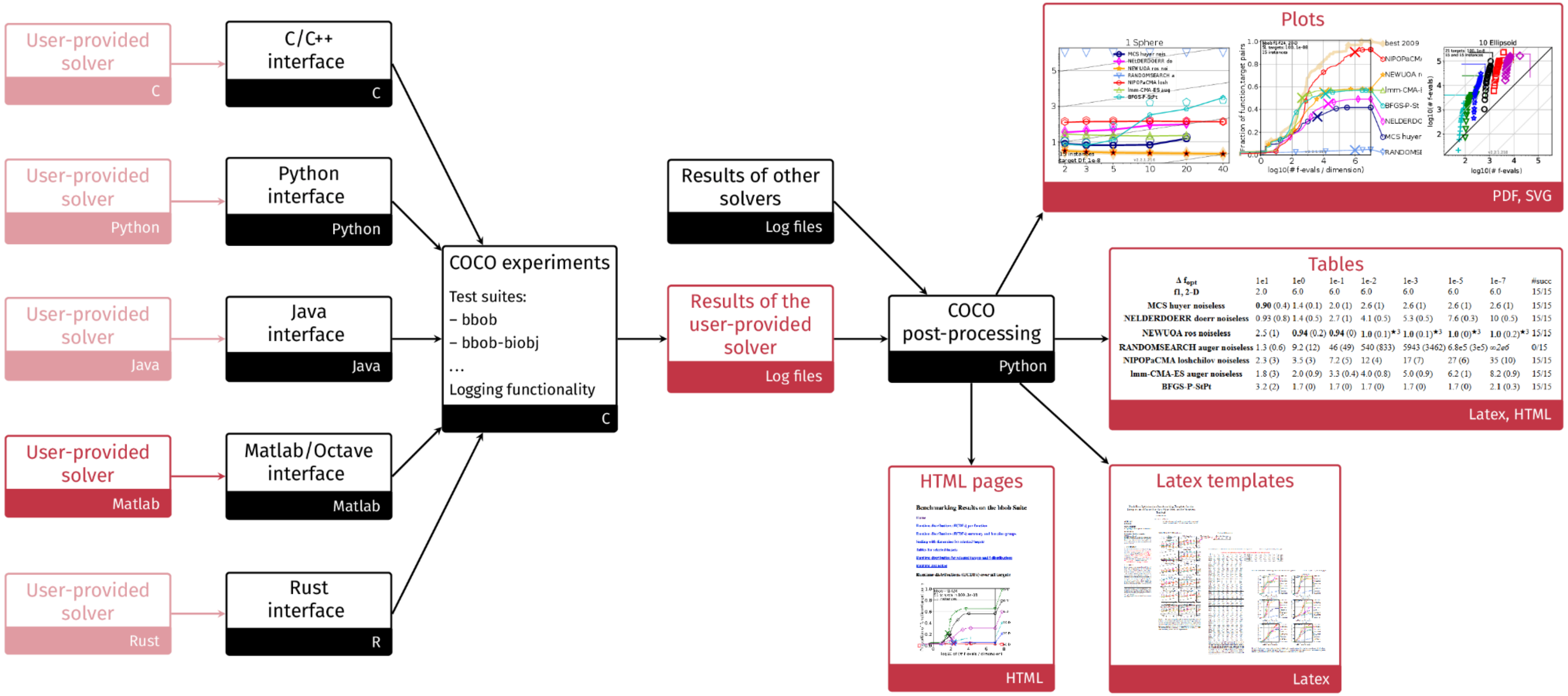
COCO: COmparing Continuous Optimizers

- COmparing Continuous Optimisers (COCO) provide benchmark function testbeds



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https://numbbo.github.io/coco/

Examples of Benchmark Functions in COCO

- The COCO/BBOB suite includes widely used test functions, such as:
  - Sphere
  - Rastrigin
  - Ackley
  - Rosenbrock
  - Griewank
  - Weierstrass
  - Schaffer F7
  - Bent Cigar

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Example – Sphere Function Optimisation

Objective:

- Find the set of integers $\boldsymbol{x_i}$ which maximise the function $\boldsymbol{f}$ given below.

$$\boxed{f(x) = \sum_{i=1}^{n} x_i^2}$$

for $n = 3$, $f(x) = x_1^2 + x_2^2 + x_3^2$

$$-512 < x_i \le 512$$

Fitness function

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Sphere Function Optimisation – Representation I

- There are 1024 integers in the given interval $-512 < x_i \leq 512$
- In binary representation, 1024 different integers can be represented using 10 bits

Encoding of x_i

0	:	0000000000	(-511)
1	:	0000000001	(-510)
…		…	
1023	:	1111111111	(512)

Decoding of x_i

- Convert binary # into decimal, e.g., ***xi*** = (0000000011) = 3
- Subtract 511 from that value, e.g., 3-511 = -509

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Sphere Function Optimisation – Representation II

- Function has 3 parameters: x_1, x_2, x_3
- Each parameter can be represented by 10 bits
- Chromosome consists of 30 bits
- Example:

1100101010	1100000000	0000000110
x_1	x_2	x_3

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Sphere Function Optimisation – Fitness Evaluation (Decoding)

1100101010 1100000000 0000000110

x_1 x_2 x_3

$$f(x) = x_1^2 + x_2^2 + x_3^2$$

$$x_1 = (\mathbf{810} - 511) = 299$$
$$x_2 = (\mathbf{768} - 511) = 257$$
$$x_3 = (\mathbf{6} - 511) = -505$$

Fitness: $f(x) = (299)^2 + (257)^2 + (-505)^2 = 410425$

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More on Function Optimisation

- What if x_i were real numbers in the interval $-5.12 < x_i \leq 5.12$
- Solution:
  - Use binary representation
    - with precision of 2 digits after decimal point
    - use 1024 different numbers
    - divide number by 100
  - Use other representation (e.g. real)
- What if the representation has redundancy?

e.g., $-5.40 < x_i \leq 5.40$

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Weaknesses

- Limited theoretical and mathematical analyses – this is a growing field of study
    - Schema theorem and building block hypothesis
    - Markov chain analysis
    - Chaos theory
    - Martingale theory (for convergence analysis)
    - Runtime analysis with respect to various parameter settings (e.g., expected runtime analysis based on fitness levels)
- Considered slow for online applications and even for some large offline problems
    - Speedy hardware enabling parallel processing

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Limited Theoretical and Mathematical Analyses

- Many optimization methods, particularly heuristic algorithms like genetic algorithms, lack rigorous mathematical foundations. This makes it challenging to predict their behavior in various scenarios.
- Implication for Benchmarking: Without robust theoretical backing, it’s difficult to establish clear performance metrics or guarantees about convergence to optimal solutions. This can lead to varying results across different benchmarks.

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Benchmark functions for discrete optimization

- **Hamming Distance** (general binary).
- **NK-Landscape** (rugged landscapes).
- **OneMax** (basic binary).
- **Royal Road** (GA-specific).
- **Combinatorial Problems** (TSP, QAP, Knapsack, etc.) with specific benchmarks

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Benchmark Suites for discrete optimization

- **OR-Library** for a variety of problem types.
- **DIMACS** for graph-based problems.
- **TSPLIB** and **CVRPLIB** for routing problems.
- **SATLIB** for satisfiability problems.
- **QAPLIB** for facility layout and assignment problems.
- etc.

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Summary

- There is a variety of benchmark functions used for performance comparison of search algorithms

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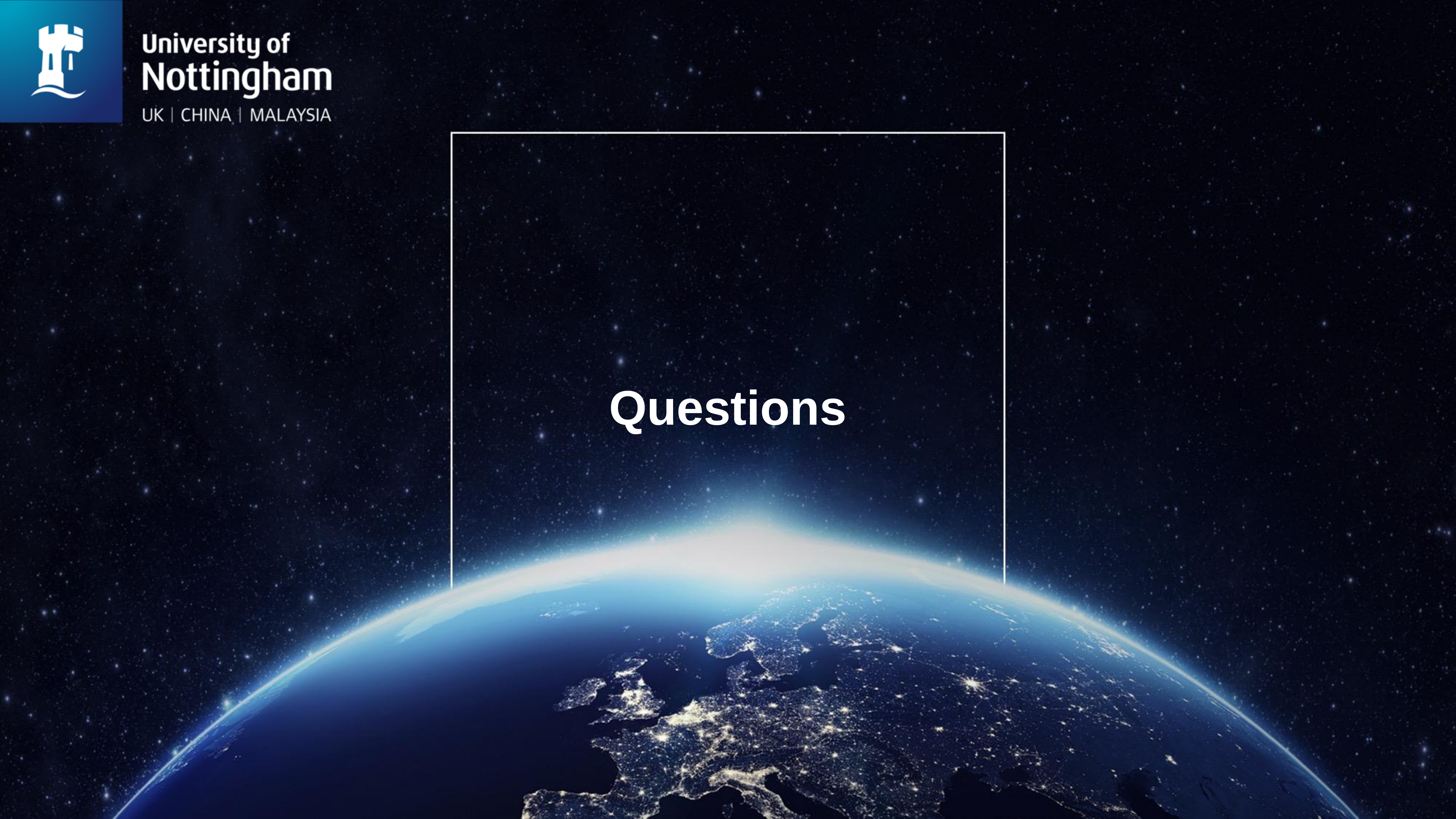

Questions

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NEXT:

Evolutional Algorithms II